An exact all-parameter verification of the bump–unwind path order in Bernoulli last-passage percolation for $n = 4$

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Abstract

We verify a finite specialization of the path-probability ordering proposed by McKeown and Nakajima for vertex last-passage percolation. On the 5×5 vertex grid with independent Bernoulli(p) weights, every genuine nonidentity bump or unwinding move strictly increases the probability of membership in the full set of maximizing paths for $0 < p < 1$. Equality holds at $p = 0, 1$. The assertion covers all 120 single-step comparisons and their 188 distinct ordered pairs in the transitive closure. Two independently written enumerators agree on every integer coefficient obtained from all 2^{23} free-vertex configurations. For 118 edges, positivity follows directly from their degree-23 nonnegative coefficient arrays. The two remaining edges have a negative coefficient, but a single exact degree elevation yields nonnegative coefficients for every edge. The exceptional edges also disprove a stronger comparison conditional on exactly three occupied free vertices. We document boundary conventions, complete certificates, small-grid hand checks, and reproducibility. The result is a modest computer-assisted finite verification; priority and independent research significance are not established.

1 Scope and relation to prior work

The question of which directed paths have greatest probability of being maximal has a geometric history; see Alberts and Cator [2, Section 8] for modal-path and path-order questions. McKeown and Nakajima [1] give two concrete path transformations and propose an increase of geodesic-membership probabilities. Their Theorem 3, equation (6), proves the Bernoulli comparison for sufficiently small positive p, with strictness for distinct related paths. The choice in their Section 5 is

$$p_n = \min\left\{\frac{1}{2N}, \frac{1}{8\binom{N}{3}}\right\}, \qquad N = (n+1)^2,$$

which gives $p_4 = 1/18400$. Our computation covers the whole parameter interval for the fixed size $n = 4$, and checks $n \leq 3$ as controls. It neither proves their general conjecture nor concerns the separate moderate-deviation issue resolved in their version 2.

We use version 2, dated 20 September 2026; the version record was checked on 29 October 2026. A bounded search of 15 query strings, author materials, and relevant backward references did not locate the same finite certificate. Forward citation coverage is incomplete: Google Scholar, ADS, and Semantic Scholar could not be fully inspected, and one author's homepage was inaccessible. Thus no priority assertion follows. The full query log, URLs, access failures, and distinctions are in `literature/audit.txt` and `literature/sources.json`. This unsigned manuscript is prepared for public archival on GitHub. It has not been submitted to a journal or sent to the cited authors. A publication-stage refresh with ten additional queries is recorded in `literature/novelty_check_20261029.txt`; priority remains unestablished.

2 Model and exact meaning of the path order

Let $V_n = \{0, \dots, n\}^2$ and let Γ_n consist of the $\binom{2n}{n}$ words containing n letters R and n letters U. A word defines the corresponding up/right path from $(0,0)$ to (n,n). For independent weights $\omega_v \sim \mathrm{Ber}(p)$ put

$$L(\gamma) = \sum_{v \in \gamma} \omega_v, \qquad L_n = \max_{\eta \in \Gamma_n} L(\eta), \qquad F_\gamma(p) = \mathbb{P}_p\{L(\gamma) = L_n\}.$$

Every tied maximizing path is counted. In particular, these probabilities do not in general sum to one; we do not select one path from a tie.

2.1 Run notation and bump moves

Write a word as its maximal constant-letter runs with signed lengths $a = (a_1, \dots, a_r)$, positive for U and negative for R. The nonzero signs alternate, the positive lengths sum to n, and the absolute negative lengths sum to n. Let $s_i = \sum_{j=1}^{i} a_j$ and $a_0 = s_0 = 0$. Thus s_i is the vertical coordinate minus the horizontal coordinate at the end of run i.

A bump has $2 \le k \le r-2$ and satisfies either $s_k \ge 0$, $a_k < 0$, or $s_k \le 0$, $a_k > 0$. Its replacement is

$$(a_1, \dots, a_{k-2},\ a_{k-1} + a_{k+1},\ a_k + a_{k+2},\ a_{k+3}, \dots, a_r).$$

Equivalently, reverse the order of the letters between corner times t_{k-1} and t_{k+1}, where $t_i = \sum_{j=1}^{i} |a_j|$. The blocks on either side merge with the resulting blocks, reducing the number of runs by two. The weak inequalities permit diagonal contact. The putative index $k = 1$ cannot satisfy the sign condition because $s_1 = a_1$.

2.2 Unwinding moves and degenerate indices

A genuine unwind has $0 \le l$, $l + 2 \le k \le r - 1$, and a sign $\varepsilon \in \{-1, 1\}$ such that

$$\varepsilon s_k > 0, \qquad \varepsilon s_i < 0 \quad (l < i < k), \qquad l = 0 \text{ or } \varepsilon s_l > 0.$$

Reverse the order of all letters between times t_l and t_k, keeping each letter's direction. For $l \ge 1$ the resulting signed runs are

$$(a_1, \dots, a_{l-1},\ a_l + a_k,\ a_{k-1}, \dots, a_{l+2},\ a_{k+1} + a_{l+1},\ a_{k+2}, \dots, a_r).$$

For $l = 0$, the prefix is absent and the first displayed merged term is simply a_k. This case reverses an initial segment, and decreases the number of runs by one; an interior unwind decreases it by two.

These conventions explicitly reconcile the run display with the geometric operations in [1, Section 2, Figures 3–4]. The strict interior signs express an actual excursion to the other side of the diagonal. If one allows $k = l + 1$ in the prose inequality, its interior condition is vacuous; the geometric segment then consists of a single constant-letter run and its reversal is an identity. The merged-run display must not be applied with overlapping indices in that case. It is excluded from our nonidentity move list. This is a stated interpretation of the genuine geometric moves, not a claim about a literal overlapping-index substitution. The source's assertion that $n = 1$ has no nontrivial moves and its Lemma 9 strict two-dot gain agree with this interpretation.

The initial boundary $l = 0$ is expressly retained. There is no extra terminal boundary: $s_r = 0$ makes $k = r$ fail the strict condition, and a_{k+1} consequently exists. Diagonal contacts at interior corners fail the strict unwind condition. No additional moves are introduced by reversing a whole path. Diagonal reflection, on the other hand, is exactly the sign symmetry already included in the definitions. An independent implementation operates on corner coordinates and word segments rather than on the merged-run formulas.

Write $\gamma \prec \pi$ when π is reachable from γ by one or more such moves. The run count strictly decreases, so this is a strict partial order. The identity belongs only to its reflexive closure.

3 Finite theorem

Theorem 1 (Computer-assisted finite comparison)**.** *For $1 \le n \le 4$, and every genuine nonidentity single-step move $\gamma \longrightarrow \pi$ defined above,*

$$F_\pi(p) > F_\gamma(p) \qquad (0 < p < 1).$$

The same strict inequality holds if $\gamma \prec \pi$. For $p = 0$ or $p = 1$, every path is maximizing, so $F_\gamma(p) = 1$. For $n = 4$, each single-step difference satisfies the explicit bound

$$F_\pi(p) - F_\gamma(p) \;\ge\; 2p^2(1-p)^{22} \qquad (0 \le p \le 1).$$

Corollary 2. *For $n = 4$ and $0 < p < 1$, the only modal paths are $\mathrm{R}^4\mathrm{U}^4$ and $\mathrm{U}^4\mathrm{R}^4$. Their probabilities are equal, and exceed that of each of the other 68 paths.*

The corner statement follows because every noncorner reaches a corner and the two corners have equal probability under diagonal reflection. This does not compare every incomparable pair in a total order.

4 Exact reduction and sign certificates

Lemma 3 (Removal of common endpoints)**.** *Setting the weights at $(0,0)$ and (n,n) to zero preserves the maximizing set for every configuration.*

Proof. Each path contains both vertices, so the same sum of two weights is subtracted from every score. All equalities and inequalities are preserved. □

Put $W_n = V_n \setminus \{(0,0),(n,n)\}$ and $m = |W_n| = (n+1)^2 - 2$. For $S \subseteq W_n$ define

$$M(S) = \max_{\eta \in \Gamma_n} |S \cap \eta|, \quad C_\gamma[k] = \#\{S \subseteq W_n : |S| = k,\ |S \cap \gamma| = M(S)\}.$$

Every set S is equally likely within its cardinality layer, and its unconditional probability is $p^k(1-p)^{m-k}$. Hence, with $q = 1 - p$,

$$F_\gamma(p) = \sum_{k=0}^{m} C_\gamma[k] p^k q^{m-k}. \tag{1}$$

The $C_\gamma[k]$ are raw configuration counts. There is no extra $\binom{m}{k}$ multiplier. If one uses the normalized Bernstein basis $\binom{m}{k}p^k q^{m-k}$, its coefficient is $C_\gamma[k]/\binom{m}{k}$.

For an edge write $d_k = C_\pi[k] - C_\gamma[k]$. When all $d_k \ge 0$ and some $d_k > 0$, equation (1) gives strict positivity for $0 < p < 1$. A negative d_k does not by itself refute the comparison.

Lemma 4 (Exact degree elevation)**.** *Set $d_{-1} = d_{m+1} = 0$ and $e_k = d_k + d_{k-1}$ for $0 \le k \le m+1$. Then*

$$\sum_{k=0}^{m} d_k p^k q^{m-k} = \sum_{k=0}^{m+1} e_k p^k q^{m+1-k}.$$

If all $e_k \ge 0$ and $e_2 \ge 2$, the difference is at least $2p^2 q^{m-1}$.

Proof. Multiply the first sum by $p + q = 1$ and collect terms of the same degree. All basis terms are nonnegative on $[0,1]$ and positive in its interior, proving the claimed bound. □

For all 120 edges at $n = 4$, the saved arrays satisfy $e_k \ge 0$ and $e_2 \ge 2$. The checks are exact integer comparisons. The same condition holds for every edge at $n = 2, 3$; $n = 1$ has no edges. No root solver, floating-point sign test, or numerical grid in p is used.

4.1 The two exceptional degree-23 arrays

The only edges failing raw coefficientwise positivity at $n = 4$ are

$$\text{RUURUURR} \longrightarrow \text{UURRUURR} \quad \text{and its diagonal reflection.}$$

The first word has runs $(-1, 2, -1, 2, -2)$ and heights $s_1 = -1, s_2 = 1$. Thus $l = 0, k = 2$ is legal: reversing the prefix RUU gives UUR. Its destination has runs $(2, -2, 2, -2)$. Both edges have the identical difference array in Table 1.

k	d_k	e_k	k	d_k	e_k
0	0	0	13	2596	6225
1	0	0	14	1347	3943
2	2	2	15	536	1883
3	-2	0	16	204	740
4	40	38	17	94	298
5	67	107	18	44	138
6	328	395	19	12	56
7	864	1192	20	1	13
8	1530	2394	21	0	1
9	2832	4362	22	0	0
10	3783	6615	23	0	0
11	3948	7731	24	$0^\dagger$	0
12	3629	7577			

Table 1: Exceptional edge coefficients. The left representation has degree 23 and the elevated representation degree 24. $^\dagger d_{24} = 0$ is only a padding convention.

Here $d_3 = -2$ is offset exactly by $d_2 = 2$ upon elevation; the next coefficient becomes $e_4 = 38$. As a second elementary sign check, the first three nonzero-degree terms can be grouped as

$$p^2q^{19}(2q^2 - 2pq + 40p^2) = \tfrac{1}{2}p^2q^{19}((2q - p)^2 + 79p^2),$$

and every remaining nonzero term is positive in the interior.

Proposition 5 (Failure of a stronger fixed-cardinality comparison)**.** *Conditioning on exactly three occupied vertices in W_4, the preceding legal unwind changes geodesic-membership probability by $-2/1771$. Thus monotonicity in each fixed-cardinality ensemble is false.*

Proof. The conditional difference equals $d_3/\binom{23}{3} = -2/1771$. This conditional model is different from the independent Bernoulli model; the proposition is not a counterexample to Theorem 1. □

5 Complete enumeration and verification

5.1 Two algorithms

Free vertices are indexed in increasing y, then increasing x, omitting the two endpoints. Paths are indexed by their complete words in lexicographic order R < U.

Implementation A, `src/count_masks.cjs`, constructs a bit mask B_γ for each path's free vertices. For each integer $0 \le b < 2^m$, it computes every score popcount$(b \,\&\, B_\gamma)$, takes their maximum, and increments the count of *every* path attaining it in the popcount(b) column. Its population count uses fixed integer bit operations.

Implementation B, `src/replay_dp.js`, independently generates paths as lists of grid indices and unpacks each configuration into individual cell weights. It obtains the global maximum by the lattice recurrence

$$D(x, y) = \omega_{x,y} + \max\{D(x-1, y), D(x, y-1)\},$$

using only the available predecessor on each boundary and $D(0, 0) = 0$. Every path's weights are then added directly and compared with $D(n, n)$. This implementation was written before inspecting A's scoring code; it neither imports A's masks nor obtains the maximum from its path scores. The shared items are the mathematical model and indexing specification.

Lemma 6 (Enumeration correctness)**.** *Each algorithm computes the integers in equation* (1)*.*

Proof. The mask loop is a bijection from the integers $0, \ldots, 2^m - 1$ to subsets of the free vertices. Recursive word generation lists every legal path once. In A, intersection population counts equal occupied path-vertex counts, and maximization is over the complete path set. In B, induction on $x + y$ proves that $D(x, y)$ is the greatest path weight to (x, y); a path's separately summed score equals $D(n, n)$ if and only if it is maximal. Both algorithms increment precisely the required integer once for every successful pair (S, γ). □

At $n = 4$ the maximum score is seven, and every count is at most 2^{23}. The unsigned 8-bit scores and 32-bit counters are sufficient. All relevant JavaScript integers are exact. In A the largest product inside the population-count reduction is below 2^{53}, so the floating-number representation does not affect its integer result. No probabilities are accumulated in floating point.

5.2 Coverage and invariants

n	m	Configurations	Paths	Bumps	Unwinds	Closure pairs
1	2	4	2	0	0	0
2	7	128	6	2	2	4
3	14	16,384	20	14	10	30
4	23	8,388,608	70	76	44	188

Table 2: Complete coverage. Closure pairs exclude identities. At $n = 4$, 30 of the 44 unwinds use $l = 0$. Each listed operation has a distinct ordered source–target pair.

Both count implementations agree on all $6 + 48 + 300 + 1680 = 2034$ entries for $n = 1, 2, 3, 4$. They separately check the cardinality layers against $\binom{m}{k}$, the endpoint columns $C_\gamma[0] = C_\gamma[m] = 1$, and $0 \le C_\gamma[k] \le \binom{m}{k}$. Counts agree under both diagonal reflection and central rotation with reversal of traversal (word reversal). These symmetries are checks on probabilities, not a license to add moves.

The signed-run and geometric word implementations agree on every operation witness $(\gamma, \pi, \text{kind}, l, k)$, not just on edge totals. Every resulting word is valid and its run count decreases by the stated amount. Their complete transitive closures agree, and the only sinks are the two corner paths.

A further check directly counts incomparable vertex pairs. Let

$$I(\gamma) = \#\{(v, w) : v \in \gamma,\ w \in W_n \setminus \gamma,\ (v_x - w_x)(v_y - w_y) < 0\}.$$

Then

$$C_\gamma[1] = 2n - 1, \qquad C_\gamma[2] = \binom{2n-1}{2} + I(\gamma).$$

Indeed, either both occupied sites lie on γ, or exactly one does and the two sites cannot lie together on any directed path. If neither lies on γ, that path cannot be maximizing. The incomparable-pair calculation matches every coefficient and yields $I(\pi) - I(\gamma) > 0$ on every edge. This independently checks the two-dot strictness underlying [1, Lemma 9] without relying on its area calculation.

5.3 Small cases by hand

For $n = 1$ each path has one nonendpoint weight, and success is the event $X \geq Y$. Its probability is $1 - p + p^2$, with raw count row $(1, 1, 1)$. There is no genuine move.

For $n = 2$, label the seven free weights

$$a = \omega_{1,0},\ b = \omega_{2,0},\ c = \omega_{0,1},\ d = \omega_{1,1},\ e = \omega_{2,1},\ f = \omega_{0,2},\ g = \omega_{1,2}.$$

The path scores in order $\mathrm{RRUU}, \mathrm{RURU}, \mathrm{RUUR}, \mathrm{URRU}, \mathrm{URUR}, \mathrm{UURR}$ are

$$a + b + e, \quad a + d + e, \quad a + d + g, \quad c + d + e, \quad c + d + g, \quad c + f + g.$$

Let $G_\gamma(t) = \sum_k C_\gamma[k] t^k$, $Q = 1 + t + t^2$, and $E = 1 + t^2$. The three symmetry classes have

$$\begin{aligned} G_{\mathrm{RURU}} &= (1 + t)Q^3 - tE^2, \\ G_{\mathrm{RUUR}} &= t(1 + t)^2 Q^2 + (1 + t + 2t^2)^2, \\ G_{\mathrm{RRUU}} &= (1 + t)(1 + t^2)Q^2 - tE^2 + tH, \\ H &= 1 + 5t + 7t^2 + 9t^3 + 3t^4 + t^5. \end{aligned}$$

For the first class, the score comparisons require $d \geq b$, $e \geq g$, and $a \geq c$. The sole remaining failure has $d = b = 0$, $a = c$, $e = g$, $f = 1$, giving the subtraction tE^2. For the second, $a \geq c$, $g \geq e$, and $d + (g - e) \geq b$, $d + (a - c) \geq f$ suffice; splitting on d gives the two summands. For the corner, split the constraint $b \geq d$ into $(b, d) = (0, 0), (1, 1), (1, 0)$. The first two cases give the first two terms combined above. In the third case the remaining comparison is $c + f + g \leq a + e + 1$, yielding

$$H = \sum_{A=0}^{2} \binom{2}{A} t^A \sum_{B=0}^{\min(3, A+1)} \binom{3}{B} t^B.$$

Expansion gives

$\gamma \backslash k$	0	1	2	3	4	5	6	7
RRUU	1	3	11	13	17	8	4	1
RURU	1	3	9	11	13	8	4	1
RUUR	1	3	9	12	14	8	4	1

and symmetry gives the remaining rows. The differences from the corner are $2t^2 + 2t^3 + 4t^4$ and $2t^2 + t^3 + 3t^4$. Thus all 48 small-case entries have a separate algebraic check.

5.4 Completion of the finite proof

The complete counts, exact move coverage, and sign certificates establish every single-step assertion of Theorem 1 by Lemmas 3 and 4. Summing the positive single-step differences along a nonempty chain establishes the strict partial-order assertion. Direct evaluation of the deterministic all-zero and all-one environments gives the endpoint equalities.

The finite data, rather than an unreported search, are part of this computer-assisted proof. The independent replay is complete, not statistical. The proof still relies on correct execution of the supplied programs and ordinary arithmetic implementations; it is not a proof-assistant formalization or a human peer-reviewed theorem.

6 Reproducibility, audit, and limitations

The project includes:

- `results/counts_n4.csv` and the corresponding JSON: all 70 count rows, with 24 entries each;
- `results/moves_n4.json`: all 120 legal operations and witnesses;
- `results/certificates_n4.json`: every difference array, its degree-24 array, and all 188 strict closure pairs;
- independent count and move outputs, and reports in `review/`;
- `src/certify.py`, `reproduce.py`, and `README.md`: exact validation and complete regeneration;
- `literature/`: source/version records and bounded search log;
- `SHA256SUMS.txt`: integrity hashes for retained artifacts.

No nonstandard Python package is required. The count implementations use Node.js; the checker uses Python's standard library. The initial $n = 4$ runs used Node v24.19.0 on Windows x64: A took 1.377 wall seconds and 1.390 CPU seconds; B took 4.563 wall seconds and 4.500 CPU seconds, with approximately 40.6 MiB peak resident memory for B. Short benchmarks preceded full runs. The initial full runs used 5.890 CPU seconds combined, well inside the 7200-second limit. The supplied portable reproduction script runs workers sequentially and rejects $n > 4$. Timings are observations on this machine, not portable guarantees. Rerun timing receipts have different hashes; deterministic count arrays and certificate arrays are the mathematical comparison targets.

Independent AI agents audited move definitions, rebuilt the full count table, checked the two exceptional sign certificates, and reviewed the manuscript. These activities are distinct from human expert review. There is no conclusion for $n \geq 5$, no assertion about other weight laws, and no claim that the finite computation is new globally or has standalone publication significance. The exceptional negative fixed-cardinality coefficient explains why positivity of every original coefficient is too strong a proposed general proof strategy; one degree elevation happens to suffice here.

References

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